College 2025-2026 Trimester 1

DETERMINING RELATIONSHIP BETWEEN FLIGHT CLASS AND AIRLINE SERVICES

STAT004 – REGRESSION ANALYSIS

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I. Problem Statement

When it comes to airline, predicting passengers' satisfaction and behavior is essential in the success of the company, and brand image. However, determining passengers' choice (Business, Eco Plus, or Eco) when it comes to flight class can be hard to predict. And what are the predictors for ticket class choice based on service quality rating and flight characteristics? Although airline companies have their own records of different flights, there is no clear data-driven model we can use to determine the relationship between flight class and airline services.

I.I Project Objective

This project objective is to:

1. To build ordinal logistic regression model in predicting flight class selection.
2. To determine important factors that affects passenger decision to upgrade to the premium class.

This project will used **Logistic Regression (Ordinal)** for predicting flight class (ordinal) while using different airline services.

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II. Dataset Description

This data is from *Kagle* and using the Airline Passenger Satisfaction.

The dataset has a total of:

- N = *25,976 samples*
- The dataset contains 13 variables.

Brief Description

- This dataset is all about airline passenger segmentation and preferences.

III. Variable Identification

- Dependent Variable: Class Variable (Business, Eco Plus or Eco)
- Independent Variable:
    - Demography: Age (Integer) and Gender (Binomial)
    - Passenger Travel Behavior: Customer Type (Binomial), Type of Travel (Binomial)
    - Flight Characteristics: Flight Distance (Integer), and Departure Delay in min (Integer).
    - Service Quality (0-5 scale): Inflight Wi-Fi service, Food and drink, Seat comfort, Inflight entertainment, On-board service, Leg room service, Baggage handling, Checking service, Inflight service, and Cleanliness.

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IV. Initial Data Explorations

Missing Values

Descriptives		
	N	Missing
Age	25976	0
Online boarding	25976	0
Leg room service	25976	0
Class	25976	0
On-board service	25976	0
Checkin service	25976	0
Inflight service	25976	0
Baggage handling	25976	0
Cleanliness	25976	0
Departure Delay in Minutes	25976	0
Inflight entertainment	25976	0
Seat comfort	25976	0
Inflight wifi service	25976	0
Food and drink	25976	0

The dataset has no missing values.

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Summary Statistics

Descriptives

	N	Missing	Mean	Median	SD	Minimum	Maximum
Age	25976	0	39.62	40.00	15.14	7	85
Online boarding	25976	0	3.26	4.00	1.36	0	5
Leg room service	25976	0	3.35	4.00	1.32	0	5
Class	25976	0					
On-board service	25976	0	3.39	4.00	1.28	0	5
Checkin service	25976	0	3.31	3.00	1.27	1	5
Inflight service	25976	0	3.65	4.00	1.18	0	5
Baggage handling	25976	0	3.63	4.00	1.18	1	5
Cleanliness	25976	0	3.29	3.00	1.32	0	5
Departure Delay in Minutes	25976	0	14.31	0.00	37.42	0	1128
Inflight entertainment	25976	0	3.36	4.00	1.34	0	5
Seat comfort	25976	0	3.45	4.00	1.32	1	5
Inflight wifi service	25976	0	2.72	3.00	1.34	0	5
Food and drink	25976	0	3.22	3.00	1.33	0	5

Key Findings:

- The average departure delay in minutes is 14.31 mins.
- 9 out of 10 services falls under the average of 3 (standard range).
- Comparing the mean and median, this distribution will show a left-skewed graph (majority of data are concentrated in right side).

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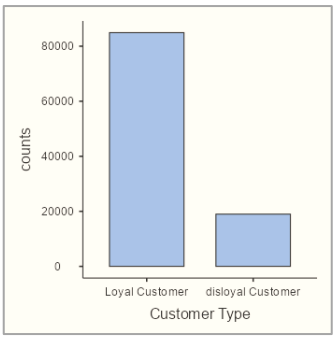
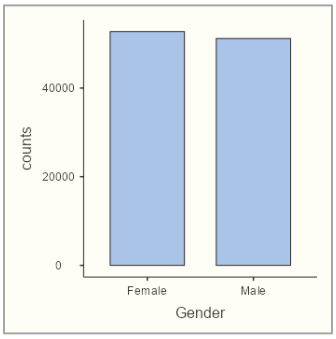
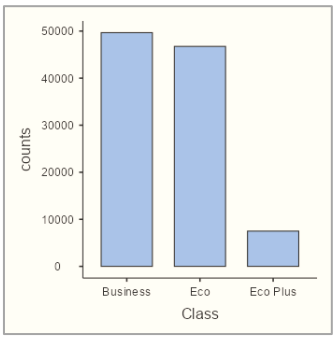
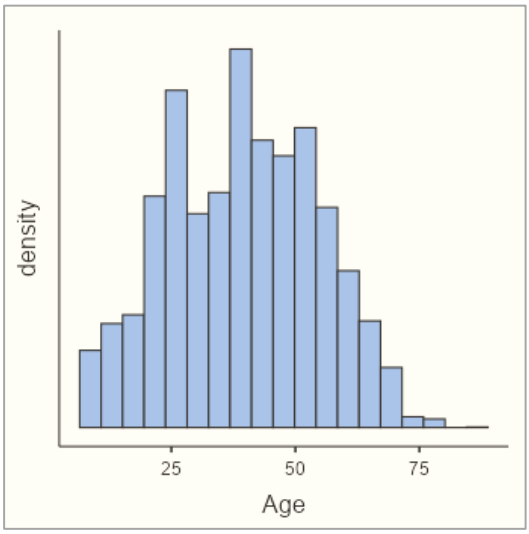
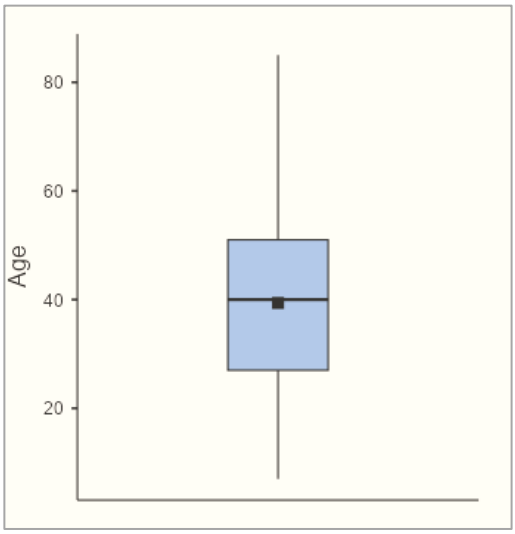
Data Visualizations











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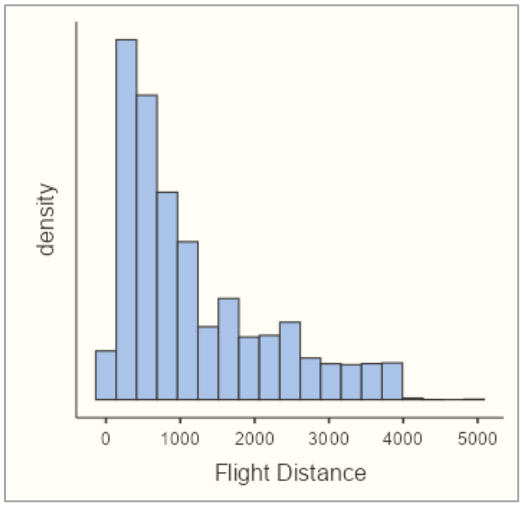

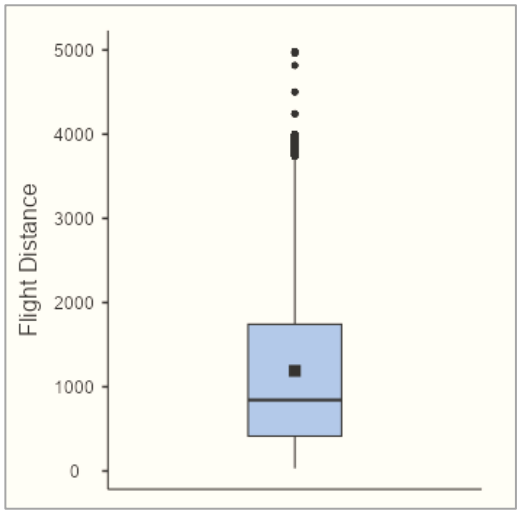

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V. Initial Hypotheses

1. H_0: There is **no significant relationship** between flight class selection and demography, travel behavior, flight characteristics, and service quality.
2. H_1: There is **significant relationship** between flight class selection and demography, travel behavior, flight characteristics, and service quality.

VI. Expected Outcome

For Information Systems, this analyzation can help optimization in airline industry revenue and determine passenger behavior using statistical method.

VII. Model Output Summary

Model Fit Measures

				Overall Model Test		
Model	Deviance	AIC	R^2_{McF}	χ^2	df	p
1	41729	41759	0.112	5270	13	< .001

Note. The dependent variable 'Class' has the following order: Business | Eco | Eco Plus

Note. Models estimated using sample size of N=25976

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The ordinal logistic regression model use was **statistically significant** based on model fit measure, **χ^2 (13) = 5270, ρ < .001**, indicating the 13 predictors when taken together, achieved better success in determining flight class. However, the model has an approximate **11.2%** of the variance in flight class **(McFadden R^2 = 0.112).** This suggest that the overall fitness prediction of the model is **weak**.

Omnibus Likelihood Ratio Tests

Predictor	χ^2	df	p
Inflight wifi service	631.85172	1	< .001
Food and drink	4.46996	1	0.034
Online boarding	1787.92687	1	< .001
Seat comfort	138.61934	1	< .001
Inflight entertainment	10.18426	1	0.001
On-board service	159.24473	1	< .001
Leg room service	400.75708	1	< .001
Inflight service	3.69965	1	0.054
Cleanliness	23.26382	1	< .001
Departure Delay in Minutes	0.00909	1	0.924
Checkin service	35.37959	1	< .001
Age	50.63219	1	< .001
Baggage handling	46.78275	1	< .001

[3]

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Based on omnibus likelihood ratio test, **two predictors** Departure Delay in Minutes (p = 924) and Inflight services (0.054) were not statistically significant predictors for flight class. While the **11 predictors** were statistically significant (**p < 0.05).**

Model Coefficients - Class

Predictor	Estimate	SE	Z	p	Odds ratio
Inflight wifi service	0.29973	0.0123	24.4420	< .001	1.349
Food and drink	0.03170	0.0150	2.1123	0.035	1.032
Online boarding	−0.52118	0.0128	−40.6857	< .001	0.594
Seat comfort	−0.17589	0.0150	−11.7605	< .001	0.839
Inflight entertainment	−0.06015	0.0189	−3.1896	0.001	0.942
On-board service	−0.16072	0.0128	−12.5992	< .001	0.852
Leg room service	−0.21651	0.0108	−19.9631	< .001	0.805
Inflight service	−0.02892	0.0150	−1.9240	0.054	0.971
Cleanliness	0.08204	0.0170	4.8158	< .001	1.085
Departure Delay in Minutes	−3.32e−5	3.48e−4	−0.0955	0.924	1.000
Checkin service	−0.06504	0.0109	−5.9469	< .001	0.937
Age	−0.00614	8.63e−4	−7.1112	< .001	0.994
Baggage handling	−0.09997	0.0146	−6.8433	< .001	0.905

[4]

Note: Using formula Percentage Change = (Odds Ratio – 1) *100%

The model coefficient table shows that several predictors significantly affected class level. **Inflight WiFi service** was a significant positive predictor of higher travel class, $b = 0.30$, $SE = 0.01$, $Z = 24.44$, $p < .001$, indicating that higher WiFi ratings increased the odds of being in a higher class by **35%** (OR = 1.35). **Food and drink quality** was also a positive predictor, $b = 0.03$, $SE = 0.02$, $Z = 2.11$, $p = .035$, though the effect was small (OR = 1.03).

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On the other hand, there were several predictors that associated with *lower* odds of being in a higher class. These included **online boarding** $b = -0.52$, $SE = 0.01$, $Z = -40.69$, $p < .001$, OR = 0.59; **seat comfort** $b = -0.18$, $SE = 0.02$, $Z = -11.76$, $p < .001$, OR = 0.84; **inflight entertainment** $b = -0.06$, $SE = 0.02$, $Z = -3.19$, $p = .001$, OR = 0.94; **on-board service** $b = -0.16$, $SE = 0.01$, $Z = -12.60$, $p < .001$, OR = 0.85; **leg room service** $b = -0.22$, $SE = 0.01$, $Z = -19.96$, $p < .001$, OR = 0.81; **check-in service** $b = -0.07$, $SE = 0.01$, $Z = -5.95$, $p < .001$, OR = 0.94; **age** $b = -0.01$, $SE = 0.00$, $Z = -7.11$, $p < .001$, OR = 0.99; and **baggage handling** $b = -0.10$, $SE = 0.01$, $Z = -6.84$, $p < .001$, OR = 0.91). These indicate that higher satisfaction with these services reduced the likelihood of being in a higher travel class, meaning these services were rated higher by lower-class passengers.

Cleanliness has a positive predictor of higher flight class, $b = 0.08$, $SE = 0.02$, $Z = 4.82$, $p < .001$, with OR = 1.09, indicating that better cleanliness ratings slightly increased the odds of being in a higher class. **Departure delay** was not a significant predictor, $b = 0.00$, $SE = 0.00$, $Z = -0.10$, $p = .924$.

Overall, the model determines inflight Wi-Fi service, food quality, and cleanliness as positive predictors of higher class, while several service ratings, including online boarding, seat comfort, and on-board service were inversely associated with higher flight class selection.

VIII. Conclusions

The analysis shows that there are many factors the influence passengers to pick flight class. Better ratings of inflight Wi-Fi, food and beverage quality and cabin cleanliness are most consistently linked to higher flight classes of travel. Others quality indicators, including seating comfort, leg room, boarding procedures and in-flight entertainment, are less likely to predictive flight classes. When combined, these results can help provide good information to use in service strategy in airlines. Airlines can find value in prioritizing high-impact amenities (Wi-Fi, food quality, cleanliness) in designing higher-class airline but also it can identify quality of comfort and benefits can also make an impact on satisfaction in the economy level cabins.

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